

Whitepaper: Intelligent Multi-Provider Auto-Routing & Resilience Engine

1. Executive Summary

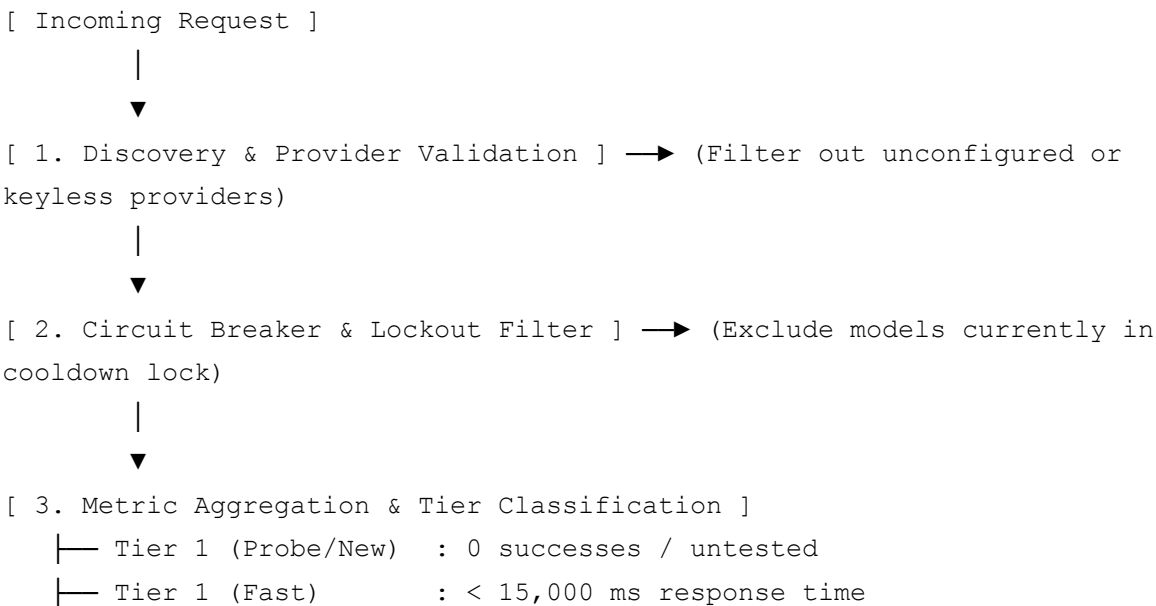
In distributed LLM applications, relying on a single provider or model creates vulnerability to rate limits (HTTP 429), regional outages (HTTP 503/500), and latency spikes. The **Auto Mode Routing Engine** is an adaptive, client-side load balancer and failure-mitigation system designed to:

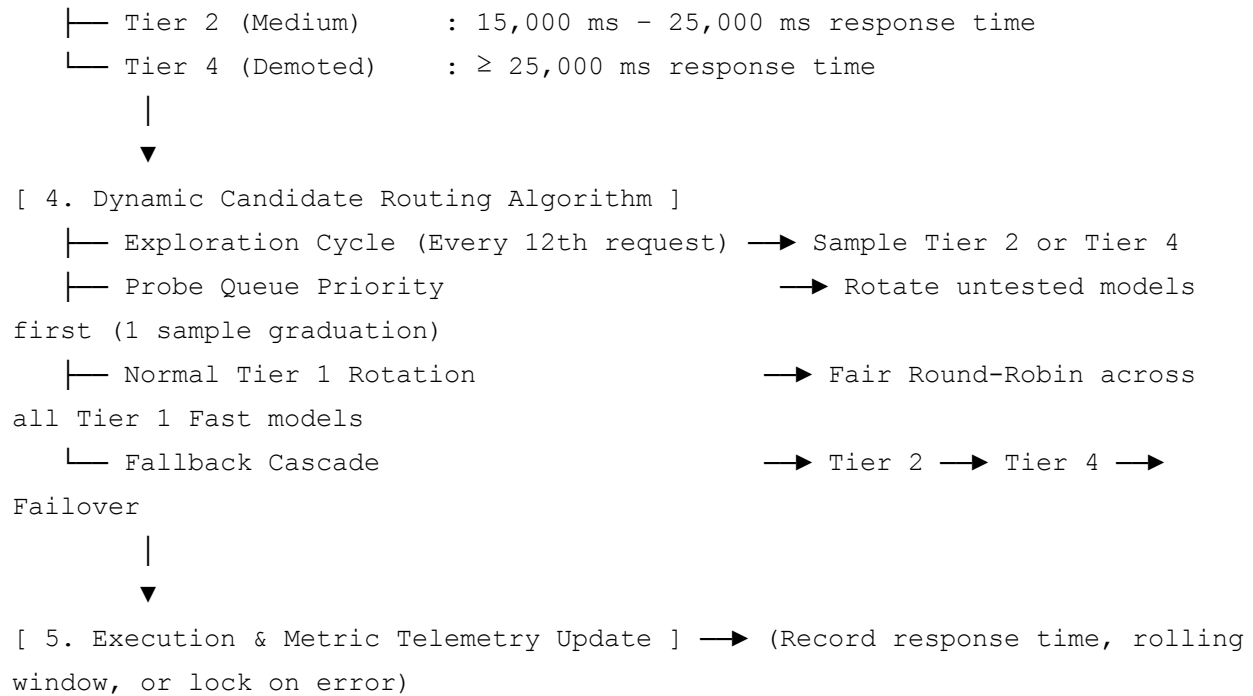
- Dynamically route requests to high-speed models across configured providers.
 - Integrate newly added models with zero cold-start starvation.
 - Ensure equal load distribution among fast models without over-stressing new additions.
 - Automatically isolate failing models using circuit breakers and re-benchmark degraded models via exploration cycles.
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2. Core Architecture & Pipeline

When an LLM operation is triggered in **Auto Mode**, the router executes a five-stage selection pipeline:

codeCode





3. Metric Tracking & Sliding Window

For every model candidate (provider, model), the engine tracks persistent telemetry in localStorage:

Telemetry State Schema

- **totalCalls**: Cumulative number of dispatched requests.
 - **totalSuccesses**: Cumulative number of successful responses (HTTP 200).
 - **recentTimesMs**: Rolling FIFO buffer of up to **10 successful response latencies**.
 - **lastResponseTimeMs**: Arithmetic mean of recentTimesMs (excluding error durations).
 - **lastTestedAt**: Epoch timestamp of the last attempt.
 - **lastError**: Timestamp and error message of the most recent failure.
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4. Performance Tier Definitions

Tier	Classification	Condition	Routing Behavior
Tier 1 (Probe/New)	High Priority (Probe)	totalSuccesses == 0 or untested	Probed first in fair round-robin until 1

			successful response is recorded.
Tier 1 (Fast)	High Priority (Active)	Effective Latency < 15s & totalSuccesses ≥ 1	Primary production pool. Rotates with equal probability in round-robin order.
Tier 2 (Medium)	Medium Speed	15s ≤ Effective Latency < 25s	Standby pool; probed periodically every 12th request for promotion.
Tier 4 (Demoted)	Slower / Degrading	Effective Latency ≥ 25s	Emergency fallback; probed periodically during exploration cycles.

5. Candidate Selection Logic

5.1 Exploration Probing (-Greedy Scheduling)

To prevent slower or recovering models from being permanently isolated, the engine maintains an exploration counter:

- If Tier 2 or Tier 4 candidates exist when the trigger fires, the engine routes the query to one of these models as an exploratory benchmark.
- If the exploratory request returns in

, the rolling average shifts and the model automatically promotes back into **Tier 1 (Fast)**.

5.2 Probe/New Model Handling

- **Untested models** (0 successful calls recorded) enter the **Tier 1 Probe Queue**.
- Probing rotates across available untested models evenly using the global rotation index ().

- **1-Sample Graduation:** As soon as a probe model records **1 successful response**, it graduates out of the probe queue and joins the standard **Tier 1 (Fast)** round-robin rotation.
- This eliminates the cold-start problem while preventing new models from being overwhelmed by cumulative call-count catching.

5.3 Normal Round-Robin Rotation

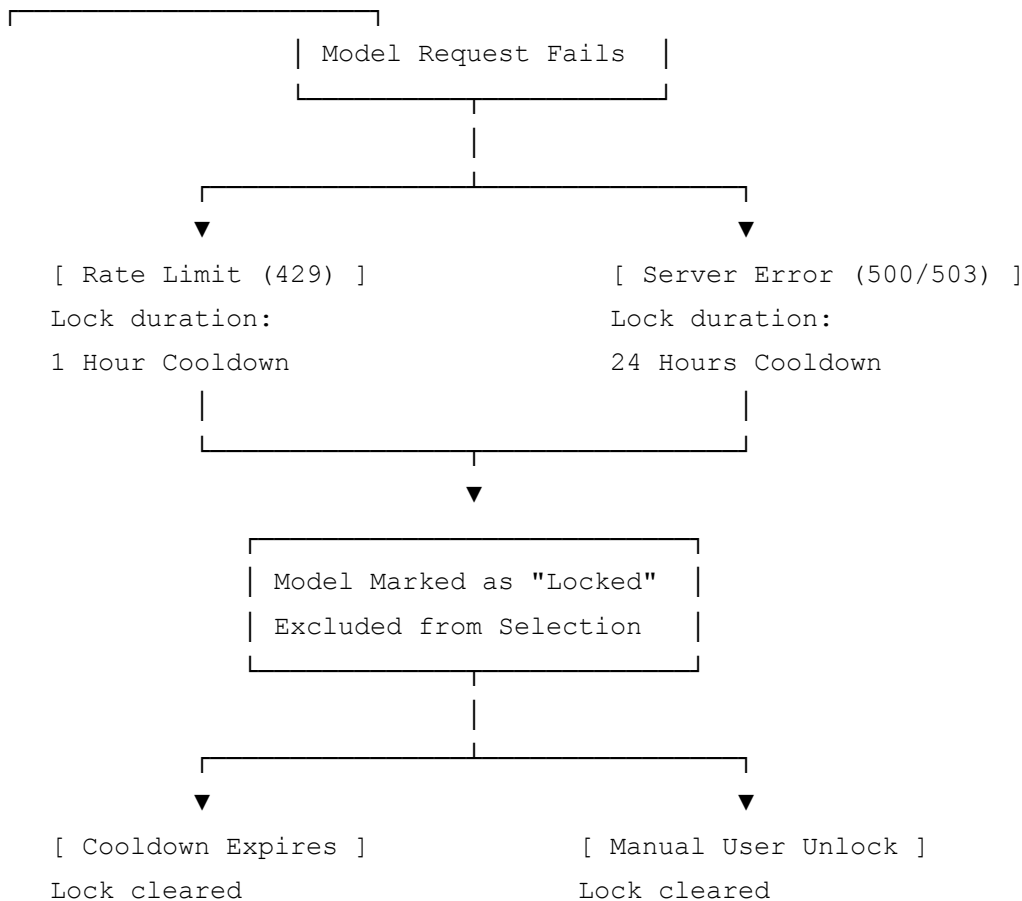
When no untested models are queued and the request is not an exploration cycle:

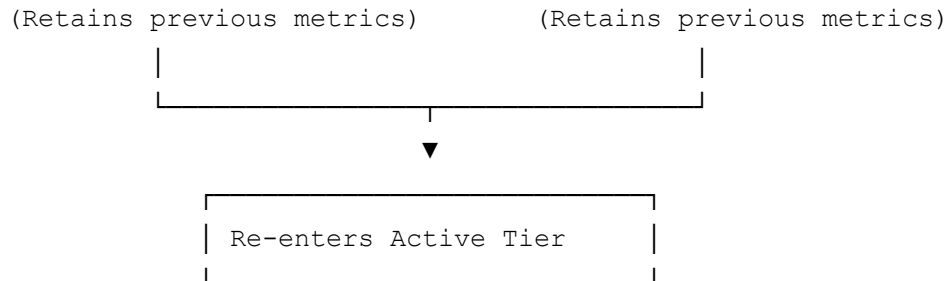
- The router cycles through all active **Tier 1 (Fast)** models evenly:

6. Circuit Breaker & Fault Recovery

When a model encounters a failure, the engine applies exponential backoff locks to prevent cascading failure loops:

codeCode





- **Rate Limit (HTTP 429):** Triggers a **1-hour lockout**.
- **Server/Quota Error (HTTP 500 / 503):** Triggers a **24-hour lockout**.
- **Manual Override:** Users can immediately clear locks via the **Model Status Modal** (Unlock Now) or reset all metrics globally (Reset All States).

7. Mathematical Summary Table

Parameter	Value	Purpose
Fast Latency Threshold		Qualifies for Tier 1 standard rotation
Medium Latency Threshold		Qualifies for Tier 2 standby pool
Exploration Frequency	Every	Re-evaluates Tier 2 / Tier 4 models for promotion
Probe Graduation Criteria		Moves model from Probe/New to active Tier 1 rotation
Rolling Metric Window		Smooths out transient network spikes
Rate Limit Lockout		Protects against quota exhaustion loops
Server Error Lockout		